shortcuts, cheating, and even more stress on the human element of the nuclear force. Indeed, from 2014 to 2016, the US Air Force was rocked by a number of scandals in the nuclear community, including massive cheating among Minuteman III missile launch operators on their nuclear certification tests.[10] As a result, highly publicized investigations forced the revamp of training and testing of missile crews. Further highlighting the pressure felt by even senior nuclear commanders, the two-star air force general in charge of all Minutemen missiles was relieved of his command of the 20th Air Force in 2013 after reports of heavy drinking and unprofessional behavior during an official trip to Russia.[11]

How the North Koreans will recruit and test their nuclear personnel, ensuring their stability and reliability, is unknown. Whether they will instill a culture of zero mistakes and how often they will test their people and critically assess their system's weak points is also unknown. Clearly fear will be a major incentive not to err, as officers who make mistakes or lose Kim Jong-un's trust are more likely to be shot than reprimanded or retired. Yet fear can easily become counterproductive, forcing more errors, especially during times of crisis. North Korea might well wind up with a system that buries mistakes (and those who make them), thereby failing to learn to do things better and more safely. That, in turn, makes even more serious mistakes far more likely, some of which could one day start a nuclear war.

V.

During the Cold War, the most harrowing specter of error hung over the command and control of nuclear weapons.[12] Command and control (C2), often paired with communications and intelligence and referred to as C3I, is the acme of nuclear surety. From storage and assembly of nuclear-capable delivery systems to clear channels of authority and decision making, to early warning and launch operations, C2 can be thought of as the pointy end of the nuclear spear.